

8 The photograph shows a wood frog.

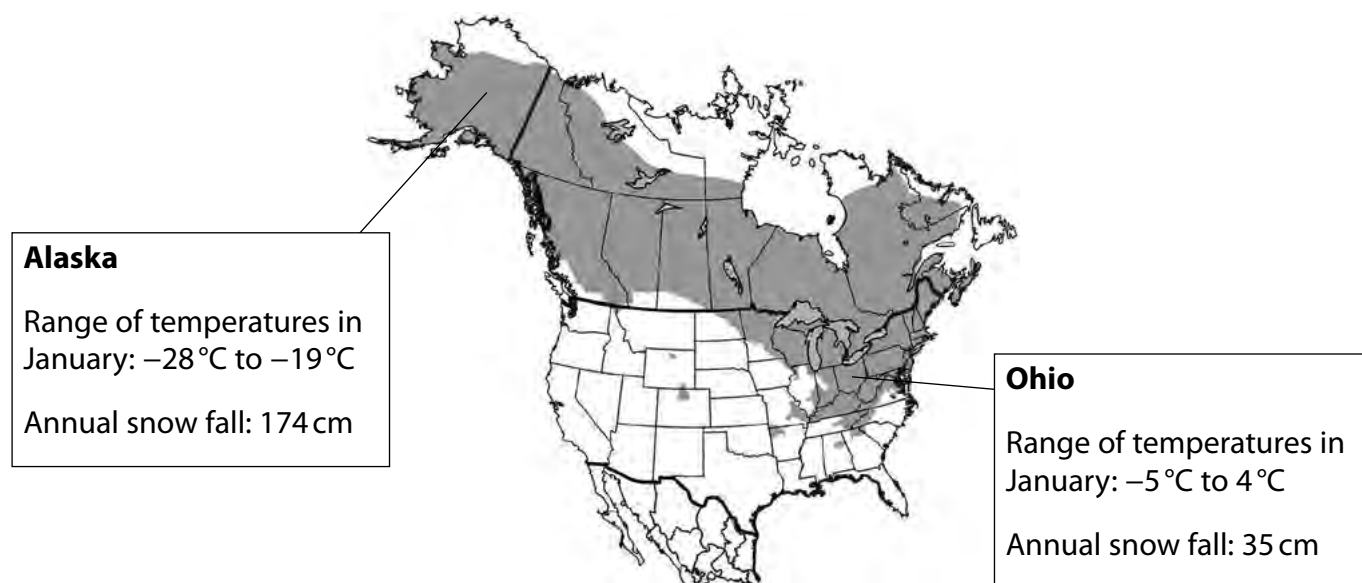


Source: http://www.borealforest.org/reptiles/wood_frog.htm

Wood frogs are found throughout North America.

The shaded areas on the map shows the distribution of wood frogs in North America.

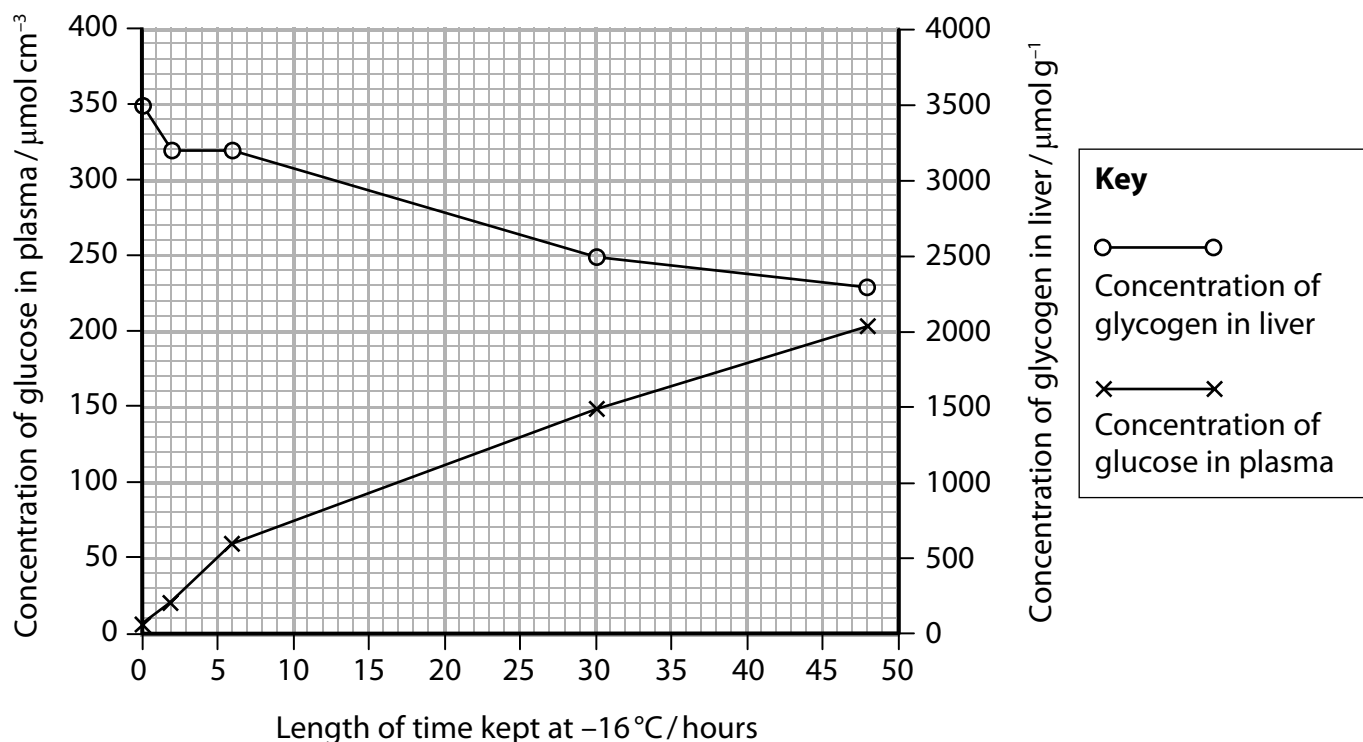
Information is also given about the climate in two areas, Alaska and Ohio.



Source: <https://answersingenesis.org/natural-selection/adaptation/the-secret-lives-of-frozen-frogs/>

- (a) Scientists kept a wood frog from Alaska at -16°C . The scientists measured the concentration of glycogen in the liver and the concentration of glucose in the plasma over a period of 48 hours.

The graph shows the results.



- (i) How many of the following statements describe glycogen?

(1)

1. The monomer is β glucose
2. It is made from two different types of polymer
3. There are 1,4 and 1,6 glycosidic bonds
4. It is insoluble

- ☐ **A** 1
- ☐ **B** 2
- ☐ **C** 3
- ☐ **D** 4

- (ii) Explain the changes in the concentration of glycogen and glucose.
Use the information in the graph to support your answer.

(2)

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- (b) The scientists found that wood frogs from Alaska had higher concentrations of glucose in their plasma than wood frogs from Ohio.

The scientists also found that the wood frogs from Alaska had higher concentrations of other solutes, such as urea, in their plasma.

The table shows the mean concentration of urea in the plasma of these frogs.
The table also shows the standard deviations.

Type of wood frogs	Mean concentration of urea in plasma / $\mu\text{mol cm}^{-3}$	Standard deviation
from Alaska	106	10
from Ohio	28	5

- (i) Calculate the percentage difference in the mean concentration of urea in the plasma of the wood frogs from Alaska compared with the wood frogs from Ohio.

(1)

Answer..... %

(ii) Explain why the data for the wood frogs from Alaska are more reliable than the data for the wood frogs from Ohio.

(2)

(iii) Explain why a high concentration of solutes in the plasma could protect the wood frogs from Alaska in very cold temperatures.

(2)

(c) Explain how these wood frogs evolved to occupy different niches in North America.

(5)

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(Total for Question 8 = 13 marks)

TOTAL FOR PAPER = 90 MARKS